

Payments Architect — Express Lane Items to Land — Q2 2026 (PAY-4026)

 [PAY-4047] ach-returns: Charge.Token & ApprovalNumber must be nullable — SqlNullValueException on real ACH rows Created: 01/Jul/26 Updated: 07/Jul/26

Status:	In Stage
Project:	Payments
Components:	None
Affects versions:	None
Fix versions:	R9.6
Parent:	Payments Architect — Express Lane Items to Land — Q2 2026

Type:	Bug	Priority:	3: Standard
Reporter:	Jason Thomas	Assignee:	Jason Thomas
Resolution:	Unresolved		
Labels:	express-lane		
Σ Remaining Estimate:	Not Specified	Remaining Estimate:	Not Specified
Σ Time Spent:	Not Specified	Time Spent:	Not Specified
Σ Original Estimate:	Not Specified	Original estimate:	Not Specified

Sub-tasks:	Key	Summary	Type	Status	Assignee
	PAY-4081	QA Task	Sub-task	To Do	Christopher Ariel Valenzuela Medina
	PAY-4082	QA: Create Test Cases	Sub-task	To Do	Christopher Ariel Valenzuela Medina
Story Points:	3				
Sprint:	PAY-Sprint 17				
QA Required:	Yes				
Grooming Status:	7-Groomed				
Pod:	PAY				

Description

Root cause

Charge.Token and Charge.ApprovalNumber (AchReturns.API/Database/Platform/Models/Charges/Charge.cs) are card-only fields — a card vault token and a card auth/approval code. They are legitimately NULL on 100% of real ACH/eCheck payment rows. The EF Core mapping in ChargeMapping.cs configured both as non-nullable string. The newer Microsoft.Data.SqlClient version pulled in transitively via the WF framework throws SqlNullValueException when materializing any real ACH row through this mapping — Data is Null. This method or property cannot be called on Null values. at SqlDataReader.GetString.

This surfaced during ACH return/NOC fixture e2e testing in Stage, where the e2e path sidestepped the bug rather than proving the fix. See design addendum bmad-wellfit #3291 §8.5 "Bug 2".

Fix

Charge.Token and Charge.ApprovalNumber changed from string to string?, and ChargeMapping.cs adds .IsRequired(false) to both property configurations.

Verification

Added a Tier 1 integration test (real SQL Server via Testcontainers, AchReturns.API.Tests/Integration/ChargeNullableCardFieldsIntegrationTests.cs) that seeds a Payments.Payments row with Token/ApprovalNumber NULL via raw ADO.NET (bypassing the EF mapping under test) and reads it back through PlatformDbContext.

- **Before fix** (files reverted to main): test failed with System.Data.SqlTypes.SqlNullValueException: Data is Null. This method or property cannot be called on Null values. at Microsoft.Data.SqlClient.SqlBuffer.get_String() / SqlDataReader.GetString — reproducing the exact production failure.
- **After fix** (fix branch restored): test passes — row materializes cleanly, Token and ApprovalNumber read back as null.

Test committed to the PR branch.

PR: <https://wellfit-technologies-inc.ghe.com/Wellfit/wellfit-payments/pull/305>

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